

BUSINESS REQUIREMENTS DOCUMENT

Patient Visit Schedule Mobile Application

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Platform	Mobile Application (Phase 1)

1. Introduction

1.1 Document Purpose

This Business Requirements Document (BRD) defines the complete functional and non-functional requirements for the Patient Visit Schedule (PVS) Mobile Application. The document serves as the primary reference for design, development, testing, and deployment teams involved in Phase 1 of the platform.

1.2 Project Overview

The Patient Visit Schedule Mobile Application is a digital platform designed to streamline the management of clinical trial visit schedules. The application enables Sponsors, CROs, SMOs, Sites, Principal Investigators (PIs), Research Teams, and Patients to coordinate, track, and manage clinical trial visit activities efficiently through a centralized mobile interface.

1.3 Scope

Phase 1 of the platform covers the following modules:

- Module 1: User Registration and Login
- Module 2: Dashboards
- Module 3: Trial / Patient Visit Schedule Management
- Module 4: Notifications
- Module 5: Calendar
- Module 6: Patient Profile & Settings
- Module 7: Chat Interface
- Module 8: Medication Reminder
- Module 9: About Trial
- Module 10: Platform Admin

The mobile application backend shall be architected to support future integration with a web-based clinical trial ecosystem platform.

1.4 Applicable Regulatory Standards

- ICH-GCP (Good Clinical Practice)
- NDCT Rules 2019 (New Drugs and Clinical Trials Rules, India)
- Indian data privacy requirements

1.5 Stakeholders

Entity	Role	Primary Access	Notes
Sponsor	Trial owner	Trial creation, site management, recruitment monitoring	No access to patient PII

CRO	Contract Research Organization	Trial creation, site management, recruitment monitoring	No access to patient PII
SMO	Site Management Organization	Site-level trial management	Can add sites
Site / Hospital	Investigational site	Patient management, trial conduct	PI and Research Team roles
Patient	Clinical trial participant	Visit schedule, medication reminders, trial info	Restricted to own data
Platform Admin	Operational oversight	User management, system monitoring	No clinical data access

MODULE 1: User Registration and Login

2. Module 1 — User Registration and Login

2.1 Objective

This module allows individual users from the clinical trial ecosystem to register on the Patient Visit Schedule mobile application, verify their identity through OTP or email verification, create login credentials, and securely access the platform.

2.2 Applicable User Entities

- Sponsor
- CRO (Contract Research Organization)
- SMO (Site Management Organization)
- Site / Hospital
- Patient

2.3 Registration Screen Flow

Screen No.	Screen Name
Screen 1	Welcome Screen
Screen 2	Select User Type / Entity Screen
Screen 3	Registration Form
Screen 4	OTP Verification Screen
Screen 5	Password Creation Screen
Screen 6	Registration Success Screen
Screen 7	Login Screen
Screen 8	Forgot Password Screen

2.4 Screen Specifications

2.4.1 Screen 1 — Welcome Screen

Purpose: To introduce the application and allow users to proceed to sign up or sign in.

- App Logo
- App Name: Patient Visit Schedule
- Button: Sign Up
- Button: Sign In
- Link: Forgot Password

2.4.2 Screen 2 — Select User Type / Entity

Purpose: To identify the type of user registering on the platform.

Field: Entity Type — Dropdown or Selection Cards:

- Sponsor
- CRO
- SMO
- Site / Hospital
- Patient

Rule: Based on the selected entity, the next registration form shall display relevant fields.

2.4.3 Screen 3 — Registration Form

The fields in the registration form vary based on the selected entity type:

#	Field	Sponsor / CRO	Site / SMO	Patient
1	Full Name	Yes	Yes	Yes
2	Designation	Yes	Yes	No
3	Email ID	Yes	Yes	Yes
4	Phone Number	Yes	Yes	Yes
5	Entity Type (auto-populated)	Yes	Yes	Yes
6	Organization Name	Yes	Yes	No
7	Organization Address	Yes	Yes	No
8	Add Sites (name and address)	No	SMO only	No
9	Role: PI / Research Team	No	Yes	No
10	Department (if PI)	No	Yes	No
11	Gender	No	No	Yes
12	Date of Birth / Age	No	No	Yes
13	Preferred Language	No	No	Yes

Organization Mapping Logic

Upon entering the organization name with the selected entity type, the system shall:

1. Search for existing organizations with similar details
2. Detect exact or similar organization names (e.g., Apollo Hospital / Apollo Hospitals / Apollo Health City may be identified as similar)
3. Suggest possible matching organizations to the user

If a similar organization exists, the user shall see two options:

- Join Existing Organization
- Enter New Organization

The system shall maintain an Organization Master Record containing:

- Unique internal Organization ID
- Standardized organization name
- Entity type
- Location details

All users associated with the same organization shall be linked to the same organization record.

Duplicate Organization Handling

- Duplicate organization detection
- Administrative organization merge capability
- Manual review by platform administrator if required

Terms and Conditions

4. The system shall display Terms and Conditions during registration.
5. Users must read and accept the Terms and Conditions before proceeding.
6. A checkbox or Accept button shall be provided.
7. The system shall not allow users to continue without acceptance.
8. The system shall store the following details for audit trail: User name, Organization name, Accepted version, Date and time of acceptance.
9. Application Admin shall be able to update Terms and Conditions.
10. If Terms and Conditions are updated, users shall be required to accept the latest version again during the next login.

Validation Rules

Before registration submission, the system shall validate:

11. Mandatory fields completed
12. Valid email format
13. Valid phone number format
14. Duplicate phone number check — display message in red: 'Phone number already exists'
15. Duplicate email check — display message in red: 'Email ID already exists'
16. Terms & Conditions accepted

2.4.4 Screen 4 — OTP Verification

Verification Options

- OTP sent to mobile number
- OTP sent to email ID

Fields

- Enter OTP
- Resend OTP

Rules

- OTP shall expire after 15 minutes
- Maximum retry attempts: 5
- If OTP fails multiple times, the account shall be temporarily locked or verification shall be paused

2.4.5 Screen 5 — Password Creation

Fields: Create Password, Confirm Password

Password Rules

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one numeric character
- At least one special character
- Password and Confirm Password must match

2.4.6 Screen 6 — Registration Success

- User account shall be created
- Success message displayed: 'Your account has been created successfully.'
- User redirected to the Sign In screen

2.4.7 Screen 7 — Login Screen

Fields

- Email ID / Phone Number
- Password (Show / Hide toggle)

Buttons

- Sign In
- Forgot Password
- Sign Up
- Remember Me (optional)

Login Rules

- Users shall be allowed to log in only after OTP verification is complete

2.4.8 Screen 8 — Forgot Password

Flow:

17. User enters registered email ID or phone number
18. OTP is sent to the registered contact
19. User verifies OTP
20. User creates a new password
21. User is returned to the login screen

2.5 Patient Registration — Two Registration Modes

2.5.1 Self-Registration

The patient can register directly through the mobile application following this flow:

22. Open the mobile application
23. Select 'Register as Patient'
24. Complete the Registration Form

25. Verify phone number through OTP
26. Create password
27. Accept Terms & Conditions and Privacy Policy
28. Login into the application
29. View the Patient Dashboard

2.5.2 Site-Initiated Registration / Invitation

When the PI or research team adds a patient to the trial (by entering phone number and email ID), an invitation link shall be sent to the patient Email ID & Phone Number. Upon clicking the link, the patient shall be redirected to the Registration Form with the email ID and phone number auto-populated.

During registration, the pre-filled fields (email ID, Phone) shall not be editable by the patient.

After registration, the system shall automatically map the invited user to the shared trial and trial access shall become active automatically.

2.5.3 Patient Consent / Declaration

During patient registration, the patient shall accept:

- Terms & Conditions
- Privacy Policy
- Consent for receiving in-app reminders
- Consent for using the app for trial visit schedule support

2.6 Business Rules

2.6.1 Role-Based Access After Login

User / Entity	Dashboard Redirected To
Sponsor	Sponsor Dashboard
CRO	CRO Dashboard
Site PI	Investigator Dashboard
Site Research Team	Research Team Dashboard
Patient	Patient Dashboard
Platform Admin	Admin Dashboard

2.6.2 Duplicate Check

The system shall not allow duplicate registration using the same:

- Email ID
- Phone Number

2.6.3 Audit Trail

The system shall record the following for every user:

- Registration date and time

- OTP verification status
- Password creation status
- Login and logout history
- Failed login attempts
- Organization mapping
- Account activation / deactivation

Each audit entry shall capture:

- Username
- Date and time
- Action performed
- IP and device information (if applicable)

2.6.4 Security Requirements

- Passwords shall be stored in encrypted format
- OTP shall not be stored as plain text
- User sessions shall expire after inactivity
- Accounts shall be locked after repeated failed login attempts
- Sensitive patient and trial-related access shall be role-based

MODULE 3: Trial / Patient Visit Schedule Management

3. Module 3 — Trial / Patient Visit Schedule Management

3.1 Module Overview

The Trial & Patient Visit Schedule Management module enables Sponsors, CROs, SMOs, Sites, Principal Investigators (PIs), Research Teams (including Clinical Research Coordinators), and Patients to manage clinical trial visit schedules digitally through the platform.

3.2 Capabilities Supported

- Trial creation
- Upload and interpretation of Patient Visit Schedule documents
- Auto-generation of structured visit schedule templates
- AI summary document with Q&A
- Site assignment
- Patient enrolment
- Auto-calculation of patient-specific visit dates
- Visit reminders and notifications
- Trial recruitment tracking

3.3 Stakeholders with Module Access

- Sponsor Users
- CRO Users
- SMO Users
- Site Users (PI, CRCs and others)
- Patients

3.4 Trial Creation

The system shall provide an 'Add Trial' option to all users (except Patients).

Upon selecting 'Add Trial,' the following fields shall be displayed:

30. Protocol ID (mandatory)
31. CTRI Number (mandatory)
32. Study Title
33. Disease / Indication (multi-select with 'Others Specify' option)
34. Study Drug Name
35. Upload Patient Visit Schedule (Supported formats: PDF, DOC, DOCX, XLS, CSV, JPEG, PNG, etc.)

Instruction displayed to user: 'Upload your Patient Visit Schedule to auto-populate the above trial fields.'

'Others Specify' Workflow

If a user selects 'Others Specify' for any field, the system shall:

36. Allow the user to enter a custom value
37. Save the value for the user and send a request to the Platform Admin
38. The Admin may Approve, Edit, or Reject the custom value
39. If approved: the value becomes available globally to all users
40. If edited: the corrected value is approved and made globally available
41. If rejected: the value remains user-specific only

This process is applicable across the entire platform wherever the 'Others Specify' option is provided.

3.5 Auto-Interpretation of Patient Visit Schedule

Upon upload of the Patient Visit Schedule document, the system shall:

- Read and interpret the uploaded schedule
- Extract visit schedule details
- Convert the extracted information into a standardized structured visit schedule format

The system shall attempt to identify:

- Visit Number
- Visit Name
- Visit Type
- Visit Gap
- Window Period
- Activities: Clinical Activities and Administrative Activities
- Other applicable fields

3.6 Standardized Visit Schedule Template

3.6.1 Template Structure

Visit Number	Visit Name	Visit Type	Visit Gap (units)
Auto-extracted	Auto-extracted	Auto-extracted	Auto-extracted

Positive Window Period	Negative Window Period	Activities	Comments
Auto-extracted	Auto-extracted	Clinical & Administrative	User editable

3.6.2 Preview, Edit & Save

Users shall be able to:

- Preview the generated template
- Edit extracted fields
- Add or remove visits
- Modify visit gaps
- Modify activities

- Add comments
- Add or remove rows and columns
- Save the final approved template
- Delete existing schedule and add a new schedule

Note: Deleted schedules shall be retained in Version History for reference.

3.6.3 Version History

The system shall maintain version history for all schedules. Version history shall capture:

- Version Number
- Date and Time
- Modified By
- Change Summary
- Previous Version Reference
- View document

3.7 Trial Saving & Management

Upon saving the trial, the system shall:

- Create the trial in the system
- Link the approved visit schedule template to the trial
- Store all trial metadata
- Generate an audit trail entry

3.8 Trial Sharing and Access Control

3.8.1 Sponsor / CRO Created Trials

If a Sponsor or CRO user creates a trial, they shall be able to:

- Share the trial internally within their organization
- Share the trial with Site users (e.g., Sponsor shares with PI)
- Add sites through single entry or bulk upload

Single Entry Fields

- Site Name
- Site Address
- PI Name
- PI Email ID
- Access Type (Patient Management — set by default)

Bulk Upload

Users shall upload site lists through PDF, Excel, DOC/DOCX, CSV, Images, etc. The system shall:

42. Read uploaded documents
43. Extract site and PI details
44. Auto-create site mappings
45. Allow users to review and edit extracted information before saving
46. Display the schedule template to added sites upon saving

3.8.2 Access Types for Internal Trial Assignment

Access Type	Description
View Access	User can view trial and schedule details only
Edit Access	User can modify trial schedule details
Patient Management Access	User can add and manage patients under the trial

Access Restriction Rules

If the trial creator is a Sponsor or CRO user:

- For Internal Trial Assignment: Disable/Hide Patient Management Access
- If the trial is shared with Sites: Disable/Hide Edit Access

3.8.3 Site-Created Trials

If a Site user creates a trial:

- The trial can only be shared with users within the same organization
- Cross-organization sharing shall not be allowed

For Internal Trial Assignment (with in the Organization members), the system shall display the following fields -

- Full Name*
- Designation
- Email ID*
- Phone Number
- Organization Name (autopopulated)
- Access Type
 - View access
 - Edit access
 - Patient Management access

3.8.4 Unregistered User Invitation Edge Case

While sharing the trial access, If the entered email or mobile number is not registered on the platform, the system shall:

- Send an invitation to the user via email
- Auto-populate the fields filled by the sender in the registration form (except email ID and Organization Name — editable by the invitee)
- To change organisation name (in case of any spelling errors), the user has to request platform admin
- Upon registration, automatically map the invited user to the shared trial
- Activate trial access automatically

3.9 Site User Functionalities

3.9.1 Create Independent Trials

- Create and manage their own active trials
- Upload their own patient visit schedules
- Manage site-level patients

3.9.2 Receive Sponsor / CRO Templates

- Receive trial schedule templates shared by Sponsors / CROs
- Access assigned trials
- Manage patients under assigned trials
- Share trials with other users (research team) within the organization

3.9.3 Patient Management — Add Patients

Fields required to add a patient:

- Subject Number / ID
- Subject Initials
- Phone Number
- Email ID
- Preferred Language for the Patient
- Screening Date OR Randomization Date

3.9.4 Single & Bulk Patient Upload

- Single patient entry
- Bulk upload through Excel, PDF, DOC, etc.

3.9.5 Auto Generation of Patient Visit Dates

Upon entering the Screening Date or Randomization Date, the system shall:

- Automatically calculate all patient visit dates
- Apply visit gaps and window periods
- Generate patient-specific visit calendars with the following fields
 - Visit Number
 - Type of Visit (Hospital Visit/ Telephonic Visit/ Home Visit)
 - Positive Window Period (Date)
 - Actual Visit Date
 - Negative Window Period (Date)
 - Visit Completion Date (manual entry)
 - Status of the Visit (Manual entry) - Screened/Screen Failure/ Follow-up/Withdrawn/Drop out/Completed
 - Entry Details

3.10 Trial Recruitment & Status Tracking

3.10.1 Sponsor / CRO View

3.10.1.1 Cumulative Recruitment Status of all site associated with a trial

- Total Screened
- Screen Failures
- Randomized
- Withdrawn
- Dropouts
- Completed
- Follow-up

3.10.1.2 Site-wise Trial Status

The Sponsor / CRO shall be able to view individual site-level performance metrics.

3.10.1.3 Patient Privacy & Access Control

Sponsor / CRO users shall NOT have access to:

- Patient names
- Full patient identifiers
- Contact details
- Sensitive medical records

Sponsor / CRO users shall only have access to aggregated recruitment and trial status metrics.

3.10.2 Site Users

The Site Users (PI & Research Team) shall be able to view the associated-trials status

- Protocol ID
- Disease/Indication
- Sponsor Name
- PI Name
- Total Screened
- Screen Failures
- Randomized
- Withdrawn
- Dropouts
- Completed
- Follow-up

3.11 Audit Trail for Trial Management

The system shall maintain audit trails for:

- Trial creation
- Schedule uploads
- Template modifications
- Site assignments
- Patient additions
- Visit modifications
- Notification activities

- Status changes

Audit logs shall capture: Username, Timestamp, Activity performed, Previous value, Updated value.

MODULE 4: Notifications

4. Module 4 — Notifications

4.1 Objective

The Notification Module provides real-time in-app reminders and alerts to users involved in patient visit management within the Patient Visit Schedule mobile application.

4.2 Notification Capabilities

- In-app notifications only (Phase 1)
- Notification history
- Read / Unread notifications
- Role-based notifications
- Patient visit reminders
- Medication reminders
- Trial sharing notifications
- Visit schedule modification notifications
- Navigation to related activity screens

4.3 Notification Section

4.3.1 Notification Channels

Notifications shall be displayed inside the mobile application (in app notifications) under the Notification Section.

Users shall be able to:

- View Notifications
- Open related activity
- Mark notifications as read/unread
- View notification timestamp
- Delete notifications

4.3.2 Notification Center

The mobile application shall include a dedicated:

- Notification Icon
- Notification Screen list

4.3.3 Notification Display Fields

- Notification Title
- Notification Message
- Sender Details
- Date and Timestamp

- Read / Unread Status

4.4 Notification Categories

Category	App Type	Description
Patient Visits	Mobile (Phase 1) + Web (future)	Visit reminders and visit updates
Medication Reminders	Mobile (Phase 1) + Web (future)	Reminders to patients about medicines
Trial Notifications	Mobile (Phase 1) + Web (future)	Trial sharing and trial updates
Meetings	Web (future)	Meeting scheduling and reminders
General	Web (future)	General system notifications

Initially, the platform shall support the following notification categories for mobile application:

- Visit Notifications (Patient visit reminders and updates)
- Medication Reminders (Reminders to Patients about Medicines)
- Trial Notification (Trial assignment and sharing)

4.5 Notification Filters

- All
- Read
- Unread

4.6 Notification Delivery

Phase 1: In-app notifications only.

Future versions may additionally support WhatsApp notifications.

4.7 Notification Trigger Rules

4.7.1 Patient Visit — Screening Visit

Trigger: When PI / Research Team enters the Baseline (screening) date of a subject.

To Patient

Message: Confirmation that the patient has been registered for the trial and information about the next visit (visit type, date, window period, site, PI details).

To PI / Research Team / SMO Admin

Message: Alert that the patient has been screened and a request to update the visit status in 'Patient Visit Details.'

4.7.2 Patient Eligible (Screen Pass)

Trigger: When PI / Research Team updates screening status as 'Screen Pass' in Patient Visit Details.

To Patient

Message: Confirmation of eligibility, next visit type, date, window period, site, and PI information.

To PI / Research Team / SMO Admin

Message: Notification that the patient is eligible and the next visit is scheduled, along with a request to update the status on the day of the visit.

4.7.3 Visit Reminders

Recipients: Patients, PI, PI Delegates, SMO / Site Admin.

Recipient	1 Week Before	1 Day Before	Day of Visit
Patient	Yes	Yes	Yes (with time)
PI / Delegates	Assignment only	Yes (Notification)	Yes (Notification)
SMO / Site Admin	Assignment only	Yes (Notification)	Yes (Notification)

4.7.4 Screen Failure Notification

Trigger: When PI / Research Team marks a patient as 'Screen Failure.'

To Patient

Message: Regret message informing the patient they are not eligible for the trial, along with the reason for screen failure.

Reasons for Screen Failure

- Not meeting Inclusion / Exclusion Criteria
- Laboratory Test Abnormalities
- Other comorbidities
- Non-compliance with Study Protocol
- Inadequate Organ Function
- Psychological Conditions
- Risk Factors for Adverse Events
- Previous Participation in a Clinical Trial
- Pregnancy or Breastfeeding
- Substance Abuse
- Inability to Provide Informed Consent
- Socioeconomic or Logistical Issues
- Baseline Health Status
- Others Specify

4.7.5 Overdue Visit Status Notification

Trigger: PI / PI Delegates fail to update the visit status by the due date.

Frequency: Notification for 3 days following the due date, then weekly reminders to PI / PI Delegates and SMO / Site Admin.

4.7.6 Subsequent Visit Completion Notification

Trigger: Visit status updated as 'Completed' in Patient Visit Details.

Recipients: Patient, PI / Delegates, SMO / Site Admin.

Applicable to all visits subsequent to the screening visit but not to the final study completion visit.

4.7.7 Patient Withdrawal Notification

Trigger: Visit status updated as 'Withdrawn.'

Message: Regret message to patient with reason for withdrawal. Alert to PI and SMO / Site Admin.

4.7.8 Patient Dropout Notification

Trigger: Visit status updated as 'Drop Out.'

Message: Alert to PI and SMO / Site Admin with reason for dropout (no patient message specified).

4.7.9 Study Completion Notification

Trigger: Last visit status updated as 'Completed.'

To Patient

Message: Grateful message acknowledging the patient's completion of the entire trial.

To PI / Research Team / SMO Admin

Message: Confirmation that the patient has completed the trial.

4.7.10 Medication Reminders

Trigger: Based on the medication schedule template and baseline date entered by the PI / Research Team.

Recipient: Patient

Message: Medication name, dosage, timing, and instructions to confirm intake after taking the medication.

4.8 Notification Screen Options

- ALL — View all notifications
- READ — View read notifications (displayed in light blue)
- UNREAD — View unread notifications
- DELETE — Select individual or all notifications to delete; deleted notifications can be restored from the bin before permanent deletion
- STOP — Stops all further alerts related to a specific activity. A confirmation hover message shall be displayed: 'Selecting the STOP option will stop all further message alerts related to this activity.'

4.9 Role-Based Notification Visibility

- Patients shall see only their own visit notifications
- PI and Research Staff shall see notifications for all patients across their assigned trials
- Notifications are displayed based on: Entity Type, Role, Trial Access, and Patient Association

4.10 Notification History

- Notification title

- Message
- Timestamp
- Read / Unread status

Older notifications may later be archived automatically.

4.11 Audit Trail — Notifications

- Notification generated
- Notification viewed
- Notification deleted
- Read status updated

Each log entry captures: Username, Date & time, Notification type, Related activity.

Note:

The Notification requirements are prepared specifically for the Patient Visit Schedule Mobile Application (Phase 1).

The detailed web application architecture and advanced workflows are available in the document “[ASSIGNMENTS AND NOTIFICATIONS \(31/12/2024\)](#)”

If any notification-related points, workflows, timelines, categories, or message formats are not explicitly covered in this mobile application requirement document, they shall be referred from and governed by the [ASSIGNMENTS AND NOTIFICATIONS \(31/12/2024\)](#)” document.

MODULE 5: Calendar

5. Module 5 — Calendar

5.1 Objective

The Calendar Module provides a centralized calendar view for managing and tracking patient visits, trial-related activities, and meetings within the Patient Visit Schedule mobile application.

The Calendar Module shall support:

- Patient visit display
- Meeting schedules (future)
- Site Visits (future)
- Day/Week/Month calendar views
- Role-based calendar visibility

The calendar module in the current phase is focused mainly on:

- Patient visits

Advanced scheduling workflows (Site Visits, full meeting management) may be introduced in later phases.

5.3 Calendar Access

The Calendar module shall be accessible from:

- Dashboard
- Menu / Navigation

5.4 Calendar Views

View Type	Description	Applies To
Day View	Activities for the selected day	All users
Week View	Activities for the selected week	All users
Month View	Activities for the selected month	All users

5.5 Visit Details in Calendar

5.5.1 For PI / Research Team

Each visit in the calendar shall display -

- Subject ID

- Protocol ID
- Visit Number
- Visit Name
- Visit Date
- Window Period
- Visit Status (updating a visit status automatically updates the calendar display)
- Site Name
- PI Name

5.5.2 For Patients

Each visit in the calendar shall display -

- Protocol ID
- Visit Number
- Visit Name
- Visit Date
- Window Period
- Visit Status (updating a visit status automatically updates the calendar display)
- Site Name
- PI Name

5.6 Role-Based Calendar Visibility

- Visibility is based on: Entity type, Trial access, and Patient assignment

Example:

- Patients shall see only their own visits
- Site users shall see assigned trials and related patients

NOTE:

The Calendar requirements are prepared specifically for the Patient Visit Schedule Mobile Application (Phase 1).

The detailed web application architecture and advanced workflows are available in the document [“CALENDAR REQUIREMENTS”](#)

If any Calendar-related points, workflows, timelines, categories, or message formats are not explicitly covered in this mobile application requirement document, they shall be referred from and governed by the [“CALENDAR REQUIREMENTS”](#) document.

MODULE 6: Patient Profile & Settings

Module 6 — Patient Profile & Settings

6.1 Objective

The Profile & Settings module allows all registered users on the Patient Visit Schedule mobile application to manage their profile information, update passwords, view Terms & Conditions, access help and support, and securely log out.

This module is applicable to all user entities:

- Sponsor
- CRO (Contract Research Organization)
- SMO (Site Management Organization)
- Site / Hospital (PI and Research Team)
- Patient
- Platform Admin

The profile fields displayed and the editability of those fields vary based on the user entity and role.

6.2 Access

- Bottom navigation menu
- Side menu or Profile icon

6.3 Profile Fields

The following table defines the profile fields applicable to each user entity:

#	Field	Sponsor / CRO Users	Site / SMO Users	Patient	Platform Admin
1	Profile Photo	Yes	Yes	Yes	Yes
2	Full Name	Yes	Yes	Yes	Yes
3	Designation	Yes	Yes	No	Yes
4	Email ID	Yes	Yes	Yes	Yes
5	Phone Number	Yes	Yes	Yes	Yes
6	Entity Type	Read only	Read only	Read only	No
7	Organization Name	Yes	Yes	No	No
8	Organization Address	Yes	Yes	No	No
9	Add Sites (name and address)	No	SMO only	No	No

10	Role: PI / Research Team	No	Yes	No	No
11	Department (if PI)	No	Yes	No	No
12	Gender	No	No	Yes	No
13	Date of Birth / Age	No	No	Yes	No
14	Preferred Language	No	No	Yes	No

Editability Rules:

- **Profile Photo:** All users may add, update, or remove their profile photo.
- **Email ID and Phone Number:** Editable upon OTP verification. If a trial is active and the user is a Patient, the system shall notify relevant stakeholders (PI & Research Team) of the contact update.
- **Entity Type and Role:** These fields are system-assigned at registration and are read-only. Changes require Platform Admin intervention.
- **Organization Name:** For Sponsor, CRO, SMO, and Site users, corrections to organization name shall be submitted to the Platform Admin for review and update.

6.4 Password Management

Fields: Current password, New password, Confirm password

Password Rules: Minimum 8 characters, uppercase, lowercase, numeric, and special character required.

6.5 Terms and Conditions

All users shall be able to view the current Terms and Conditions within the application.

- Patients shall additionally be able to view the Privacy Policy and the consent declarations accepted during registration.
- If Terms and Conditions are updated, users shall be required to accept the latest version upon the next login..

6.6 Help and Support

- FAQ / Help section
- Contact support
- Report issue

6.7 Session Management

This section is applicable to all user entities.

- The system shall maintain secure login sessions for all users.
- Sessions shall automatically expire after a configurable period of inactivity.
- Re-login shall be required after a password reset.
- Active sessions shall be terminated upon logout.

6.8 Audit Trail — Profile & Settings

- Profile updates
- Password changes
- Login / Logout activity
- Session Expiry events
- Contact information updates (with verification status)

Each audit entry shall capture: Username, Entity Type, Field Updated, Previous Value (masked where applicable), Updated Value, Date and Time.

6.9 Security & Privacy

- Each user can access only their own profile.
- Sensitive information such as passwords shall be stored in encrypted format.
- Contact details (Email ID and Phone Number) shall be editable only after OTP verification.
- Patient profile information shall not be visible to Sponsor or CRO users.
- Role-based access control shall be enforced.
- Secure authentication shall be maintained at all times.

MODULE 7: My Visits (Patient)

7. Module 7 — My Visits (Patient)

7.1 Objective

The My Visits module allows patients to view and track all their clinical trial visits in a simple mobile interface.

The module shall display:

- Upcoming visit
- All visit list
- Visit details
- Visit status
- Visit reminders
- Visit instructions
- Visit window period
- Visit type

7.2 Access Points

Patients shall access this module from:

- Bottom menu: My Visits
- Dashboard — Upcoming Visit card
- Notification click-through
- Calendar visit entry

7.3 Screen Layout

A list of associated trials (including past trials) shall be displayed. Upon selecting a trial, the screen shall consist of three panels:

7.3.1 Panel 1 — Trial Details

- Protocol ID
- Study Title
- Disease / Indication
- Drug Name
- Global / Domestic
- Hospital Name
- PI Name

7.3.2 Panel 2 — Upcoming Visit

- Visit Number
- Visit Name

- Visit Type (Hospital / Telephonic / Home)
- Scheduled Date and Time
- Visit Window
- Hospital Name
- PI Name
- Visit Status
- Instructions

7.3.3 Panel 3 — All Visits

- Visit Number
- Visit Name
- Visit Type (Hospital / Telephonic / Home)
- Scheduled Date and Time
- Visit Window
- Visit Status

Note: The fields displayed may vary based on the uploaded visit schedule.

MODULE 8: Chat Interface

8. Module 8 — Chat Interface

8.1 Objective

The Chat Interface module enables secure communication between patients, Principal Investigators (PI), and the Research Team to improve patient engagement, visit coordination, medication communication, and trial-related support.

8.2 Communication Participants

- Patient
- Principal Investigator (PI)
- Research Team

8.3 Chat Features

Feature	Description
Group Chat	Supports multi-party communication within a trial group
Text Messaging	Standard text-based messaging
Voice Messaging	Voice message recording and playback
Voice-to-Text Conversion	Converts voice messages to readable text
Document Upload & Sharing	Supports sharing of documents within the chat
In-app Chat Notifications	Alerts for new messages
Chat History	Full message and document history maintained
Message Timestamps	Date and time stamp on every message
Role-Based Access	Access restricted by entity and role

8.4 Message Display Fields

- Sender Name
- Sender Role
- Message Content
- Date and Time Stamp
- Attachment Indicator (if document attached)

8.5 Chat History

The system shall maintain full message history, voice message history, and shared document history. Chat history shall remain linked to the trial, Patient / Subject ID, and site.

8.6 Access Points

- Dashboard
- My Visits
- Notifications
- Patient Profile
- Dedicated Chat menu

8.7 Security & Privacy

- Secure message transmission
- Encrypted storage
- Role-based access control
- Secure document handling
- Patients can access only their own trial group
- Unauthorized users cannot access chats
- Sensitive patient or trial information shared only with authorized users

8.8 Audit Trail — Chat

- Message sent
- Voice message uploaded
- Document shared
- Message viewed
- Message deleted
- User added / removed from group

Each log captures: Username, Date & time, Activity type.

MODULE 9: Medication Reminder

9. Module 9 — Medication Reminder

9.1 Objective

The Medication Reminder module helps clinical trial patients adhere to protocol-defined medication schedules by providing timely reminders and allowing patients to confirm medication intake through the mobile application.

9.2 Capabilities

- Medication schedule creation
- Medication reminders
- Medication intake confirmation
- Medication status tracking
- Reminder notifications
- Date and time stamp capture
- Medication history
- Role-based visibility

9.3 Medication Schedule Creation

Authorized research team users shall be able to:

- Upload a medication schedule linked to patient visits / trial schedule, OR
- Allow the system to auto-calculate from the patient visit schedule / Protocol, OR
- Manually enter medication details

Manual entry fields:

- Medication Name
- Dosage
- Route of Administration
- Frequency
- Scheduled Time
- Start Date
- End Date
- Instructions

9.4 Reminder Notifications

9.4.1 Reminder Notification Message (refer to Medication reminders section in Notification)

9.4.1 Reminder Timing (It is based on the Frequency entered by the PI/ Research team or captured from Medication chart/Protocol/Patient Visit Schedule)

9.5 Patient Medication Actions

Upon receiving a medication reminder, the patient shall be able to select:

- Medicine Taken
- Medicine Not Taken
- Medicine Skipped
- Remind Me Later

9.6 Medication Confirmation Capture

When a patient updates medication status, the system shall capture and display the following to the PI / Research Team:

- Patient ID / Initials
- Medication Name
- Dosage
- Status: Taken / Not Taken / Skipped
- Date and Time

9.7 Access Points

- Dashboard
- Notifications
- Dedicated Medication Reminder menu
- My Visits

9.8 Security & Privacy

- Patients can access only their own medication schedule
- Site users can access all patient medication data for their assigned trials
- Unauthorized users cannot access medication records

Medication reminders are support tools and do not replace investigator instructions or medical advice

9.9 Audit Trail — Medication

- Medication reminder generated
- Reminder viewed
- Medication status updated
- Confirmation timestamps
- Medication schedule modifications

MODULE 10: About Trial

10. Module 10 — About Trial

10.1 Objective

The About Trial module provides patients with simplified, patient-friendly information about the clinical trial in which they are participating.

The module shall support:

- Display of the Trial Summary
- Visit Schedule Summary (Overview)
- Site/Research Team Contact Details

10.2 Access

- Dashboard
- Side menu / Navigation

10.3 Trial Information Source

About Trial content shall primarily be generated from:

- Patient Information Sheet (PIS)
- Trial summary details
- Site / Research team inputs

10.3.1 PIS Upload

PI / Research team / Site users shall be able to upload Patient Information Sheet documents in the following formats:

- PDF
- DOC / DOCX
- Scanned copies

10.3.2 PIS-to-Template Conversion

The system shall:

- Read the uploaded PIS document
- Extract relevant information
- Convert information into structured mobile-friendly sections using OCR, text extraction, and AI-assisted summarization

10.3.3 About Trial Sections

Section	Description
Trial Title	Name / title of the study
Trial Purpose	Why the study is being conducted
Trial Summary	Simplified study overview
Study Duration	Expected participation duration
Visit Schedule	Overview of planned visits
Patient Responsibilities	What the patient needs to do
Medication Information	Trial medication details and instructions
Risks & Discomforts	Simplified risks
Benefits	Potential benefits
Emergency Contact	Site / Research team contact
PI Details	Investigator information
Site / Hospital Details	Trial site information
Withdrawal Information	Patient withdrawal rights

10.4 Language Support

The About Trial module shall support multiple languages based on the Preferred Language selected by the patient during registration.

10.5 Trial Updates

If trial-related patient information is updated, the About Trial section shall be updated and relevant notifications may be generated.

10.6 Audit Trail — About Trial

- PIS upload
- Trial content update
- Patient access / view history
- Download activity

Audit logs shall include:

- a) User
- b) Activity
- c) Date & time

MODULE 11: Platform Admin

11. Module 11 — Platform Admin

11.1 Overview

The Platform Admin manages and monitors the overall functioning of the Patient Visit Schedule Mobile Application. The Platform Admin role is operational and technical in nature and shall not participate in clinical decision-making, patient treatment, or trial conduct activities.

11.2 Responsibilities

- Monitor platform usage
- Support users
- Manage registrations
- Handle basic security issues
- Monitor notification delivery
- Resolve operational issues
- Ensure smooth functioning of the application

The Platform Admin is not intended to access confidential clinical trial data unnecessarily.

11.3 Access Permissions

11.3.1 Platform Admin SHALL access

- View registered users
- View organizations
- View trial names and basic trial information
- View notification delivery status
- View app usage statistics
- Reset passwords
- Suspend suspicious accounts
- Resolve support issues

11.3.2 Platform Admin SHALL NOT access

- Patient medical records
- Lab reports
- Clinical data
- Chat content
- Uploaded confidential trial documents
- Audio / video communications

11.3.3 Data Masking

The system shall implement masking of sensitive information wherever applicable.

Example:

Data Field	Original Format	Masked Format
Patient Name	Divya Yadav	D*** Y***
Mobile Number	9876504582	*****4582
Email ID	divya@gmail.com	di***@gmail.com

11.4 Platform Admin Dashboard

The system shall provide a simple dashboard displaying:

- Total registered users with entity-based categorization (Sponsor Users, CRO Users, PIs, Research Team, Patients)
- Total organizations with entity-based categorization (Sponsor, CRO, SMO, Sites, Patients)
- Total trials: Active, Completed, Terminated / Suspended
- Notifications: Sent today, Failed count
- Pending support requests
- Pending activities (Others Specify approvals, Organization Names corrections, etc.)
- Data storage usage

Dashboard Filters

- Date range
- Organization type
- Trial status
- User role

11.5 User Management

11.5.1 User List View Fields

- User ID
- Profile Photo (if uploaded)
- Name
- Email
- Mobile Number
- Organization Name
- Organization Type
- User Role
- Registration Date
- Last Login Date
- Account Status

11.5.2 Search and Filter

Search by: Name, Email, Mobile Number, Organization Name.

Filter by: Organization Type, User Role, Account Status, Registration Date.

11.5.3 Account Status Types

Status	Description
Active	User can access the platform
Pending Verification	Verification is incomplete
Suspended	User access is temporarily blocked

11.5.4 Admin User Actions

The Platform Admin shall be able to:

- Activate user accounts
- Suspend user accounts
- Unlock user accounts
- Reset passwords
- Assist with login issues or account recovery

Note: All admin actions shall be recorded in audit logs.

11.5.5 Patient User Visibility

The Platform Admin shall only view limited patient information:

- Patient ID
- Patient initials / masked name
- Masked Phone Number
- Masked Email ID
- Trial association
- Account status

The Platform Admin shall NOT access:

- Medical records
- Lab reports
- Clinical notes
- Chat messages
- Uploaded medical documents

11.6 Organization Management

- View registered organizations
- Correct duplicate organization entries (merge duplicate profiles)
- Correct spelling variations and standardize organization names
- Correct organization mapping errors
- Suspend suspicious organizations
- View number of users linked to an organization
- View number of active trials linked to an organization

11.5.6 Display Fields

- Organization name
- Organization type
- Organization address
- Registered contact details
- Number of users
- Number of trials

11.7 Trial Monitoring

The Platform Admin may view trial activity at a high level for operational purposes:

- Trial title
- Trial ID
- Sponsor / CRO name
- Site name
- Trial status
- Number of patients enrolled
- Schedule version number
- Date of modification
- Modified by
- Change summary

The Platform Admin **shall NOT**

- Modify trial schedules
- Modify patient visits
- Access clinical outcomes
- Access confidential protocol sections.

11.8 Privacy and Data Protection

The system shall ensure that:

- Patient confidentiality is protected
- Clinical trial confidentiality is maintained
- Role-based access control is implemented
- Sensitive information is masked
- All admin activities are audit trailed

The platform shall align with:

- HIPAA
- GDPR
- 21 CFR Part 11
- ICH-GCP
- NDCT Rules 2019
- Indian data privacy requirements

The system shall ensure:

- Least privilege access

- Need-to-know access
- Secure password encryption
- Session timeout controls

All Platform Admin activities shall be logged and audit trailed.

11.9 Notification Monitoring

The Platform Admin shall be able to monitor:

- Visit reminder notifications
- Medication reminder notifications
- Trial sharing notifications
- Failed notifications

The Platform Admin shall be able to:

- Configure notification content
- Configure reminder timelines
- Configure system-wide notification settings

The Platform Admin shall be able to:

- Retry failed notifications
- View notification history
- Monitor delivery success/failure status
- Configure basic reminder timing settings

The Platform Admin shall not access confidential patient communication content unnecessarily.

11.10 Terms and Conditions Management

- Create Terms & Conditions versions
- Update Terms & Conditions
- Update Privacy Policy
- Activate / Deactivate versions
- View acceptance history

The system shall force users to accept the updated Terms & Conditions during next login and maintain audit trail of all acceptance events.

11.11 Master Data Management

The Platform Admin shall be able to:

- Review custom values submitted under “Others Specify”
- Approve custom values
- Edit custom values
- Reject custom values

If approved:

- value becomes globally available

If edited:

- corrected value becomes globally available

If rejected:

- value remains user-specific only

The system shall maintain audit logs for:

- Approval
- Rejection
- Editing of custom values

The system shall maintain audit logs for all approvals, rejections, and edits of custom values.

11.12 Support and Help Desk

11.12.1 Support Request Categories

- Login issues
- Password reset
- Notification issues
- Trial access problems
- App technical issues

11.12.2 Ticket Management Fields

The Platform Admin shall be able to:

- View support requests
- Respond to issues
- Resolve technical problems
- Update ticket status
- Mark tickets resolved

11.12.3 Ticket Management

- Ticket ID (unique, system-generated)
- User name
- Issue category
- Date / Time raised
- Ticket status (open/In progress/Resolved/Closed)
- Resolution notes

Ticket Status	Meaning
Open	Issue received, not yet actioned
In Progress	Admin is working on resolution
Resolved	Issue addressed
Closed	Fully closed

11.13 Audit Trail — Admin

Audit logs shall be maintained for:

- User registration
- Login / Logout
- Password reset
- Trial creation
- Trial sharing
- Patient addition
- Visit schedule updates
- Notification activities
- Admin actions

Audit logs shall capture: User / Admin ID, Action performed, Date and time, Status of action. Audit logs shall be non-editable.

11.14 Security Controls

- Suspend suspicious users
- Force logout of active sessions if required

11.15 Reports

The Platform Admin shall generate the following reports:

- Registered users report
- Organization-wise user report
- User status report
- Login activity report

Reports shall support filtering by date, organization, and trial, and shall support PDF and Excel export.

11.16 Delegation to MTB Staff

The main Platform Admin may delegate limited operational activities to authorized MTB staff including:

- User support
- Password reset assistance
- Notification monitoring
- Ticket management.
- Other Tasks (Configure Others Specify Values; Standardize Organization Names, etc)

All delegated actions shall be audit logged.

11.17 Emergency Access

11.17.1 Purpose

The system may provide restricted emergency access to the Platform Admin only under exceptional situations.

11.17.2 Permitted Situations

Emergency access may be permitted for:

- Serious technical failure investigation
- Security breach investigation
- Legal or regulatory requirement
- Data recovery activity
- Critical patient safety issue related to system malfunction
- Court / Government authority request

11.17.3 Conditions for Emergency Access

Emergency access shall require:

- Valid reason / justification entry required
- Authorized Senior Personnel approval required
- Time-limited access
- Audit logging of all activities
- Access only to the minimum required information

11.17.4 Emergency Access Audit Trail

- Admin name
- Reason for emergency access
- Date and time
- Information accessed
- Duration of access
- Actions performed

Emergency access logs shall be non-editable.

Non-Functional Requirements

12. Non-Functional Requirements

12.1 Security

- All passwords shall be stored in encrypted format
- OTP shall not be stored as plain text
- User sessions shall expire after inactivity
- Accounts shall be locked after repeated failed login attempts
- Role-based access control shall be enforced throughout the platform
- Least privilege access principle shall be applied
- Need-to-know access principle shall be applied
- Secure password encryption
- Session timeout controls

12.2 Privacy and Data Protection

- Patient confidentiality shall be protected at all times
- Clinical trial confidentiality shall be maintained
- Sensitive information shall be masked where applicable
- All admin activities shall be audit trailed
- The platform shall align with ICH-GCP, NDCT Rules 2019, and Indian data privacy requirements

12.3 Performance

- Fast dashboard loading
- Real-time notification delivery
- Responsive UI on all mobile screen sizes

12.4 Scalability & Architecture

- The mobile application backend shall be designed to support future integration with a web-based clinical trial ecosystem platform
- Scalable architecture for future enhancements
- High availability
- Basic backup and recovery support

12.5 Usability

- Mobile responsive UI
- Simple navigation
- Multi-language support based on patient's preferred language

13. Glossary

Term / Abbreviation	Definition
BRD	Business Requirements Document
CRC	Clinical Research Coordinator
CRO	Contract Research Organization
CTRI	Clinical Trials Registry — India
ICH-GCP	International Council for Harmonisation — Good Clinical Practice
MFA	Multi-Factor Authentication
NDCT	New Drugs and Clinical Trials Rules
OCR	Optical Character Recognition
OTP	One-Time Password
PI	Principal Investigator
PIS	Patient Information Sheet
PVS	Patient Visit Schedule
SMO	Site Management Organization
Subject ID	Unique identifier assigned to a clinical trial participant
Window Period	Acceptable date range before or after a scheduled visit

14. Document Control

Attribute	Details
Document Title	Business Requirements Document — Patient Visit Schedule Mobile Application
Version	1.0
Status	Draft
Phase	Phase 1 — Mobile Application
Governing Standards	ICH-GCP, NDCT Rules 2019, Indian Data Privacy Requirements
Review Cycle	As required upon change in scope or regulatory requirements
Classification	Confidential — For Authorized Use Only

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